Corpus Engineering for Fish-Species Artificial Intelligence: A Species-First, Multimodal Survey and Protocol Handbook

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Abstract

Fish-species artificial intelligence is often treated as a conventional image-classification problem, yet the observations required for reliable deployment depend strongly on species ecology and on the sensing modality. Schooling pelagics may be best represented by sonar and continuous video; cryptic benthic fishes require masks and hard background negatives; migratory salmonids require track- and passage-level splits; soniferous fishes require verified call libraries; rare species may depend on eDNA and provenance-rich opportunistic observations. This paper synthesizes corpus-construction lessons from major fish and marine datasets, fish bioacoustics, hydroacoustics, and eDNA methodology, and proposes a three-tier corpus architecture: a protected gold core, a validated supervised set, and a weakly labeled/unlabeled reservoir under a shared taxonomy and metadata plane. Because FishBase lists tens of thousands of fish species, the paper does not claim to enumerate every species. Instead it supplies a decision procedure applicable to any species and detailed worked corpus protocols for 64 representative species spanning pelagic, demersal, migratory, aquaculture, reef, cryptic, model, elasmobranch, and deep-sea archetypes. The central recommendation is to optimize for independent-event coverage, confuser diversity, provenance, leakage-resistant evaluation, and open-world uncertainty rather than raw frame count.

Keywords: fish species recognition; corpus construction; dataset design; underwater computer vision; sonar; eDNA; passive acoustics; tracking; long-tail recognition; biodiversity monitoring.

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1 Scope, terminology, and the meaning of a fish-species corpus

A machine-learning "corpus" for fish species is broader than a directory of photographs. It is a governed collection of observations, labels, metadata, negative examples, provenance, and evaluation partitions designed to support a stated biological or engineering decision. Depending on the species and deployment, the observations may be RGB images, video clips, stereo imagery, imaging sonar, scientific echosounder backscatter, passive-acoustic recordings, environmental-DNA sequences, specimen images, morphometrics, or combinations of these modalities. The core claim of this survey is that the optimal corpus is therefore *species-conditional*: the acquisition and annotation protocol should be chosen from the animal's ecology, morphology, behavior, observability, and the intended sensing platform rather than copied from a generic image-classification recipe.

FishBase listed approximately 36,700 species in its June 2026 release, which makes literal species-by-species enumeration in a single paper impractical [1]. This handbook instead provides two complementary deliverables. First, it defines a decision framework that maps any species to an acquisition, labeling, splitting, and evaluation strategy. Second, it gives detailed worked protocols for 64 representative species spanning pelagic schooling fishes, large pelagics, demersal and flatfishes, salmonids and migratory fishes, aquaculture species, reef fishes, cryptic taxa, model organisms, elasmobranchs, and deep-sea fishes. These profiles are not biological monographs; they are corpus-engineering cards designed to be adapted to the actual project.

The word "best" is used here in an operational sense. A corpus is better when a model trained on it transfers to the intended deployment domain, maintains calibrated uncertainty, exposes failure modes, supports biologically meaningful evaluation, and can be audited and extended. A larger corpus can be worse if it repeats adjacent frames from a few events, leaks sites into the test set, contains unverified labels, or over-represents easy iconic views. This principle is supported by the evolution from constrained crop datasets toward in-situ datasets such as DeepFish, OzFish, Fishnet Open Images, FishNet, FathomNet, and sonar benchmarks that explicitly emphasize environmental variation and out-of-location testing [2–5, 7, 10].

2 Evidence base: what existing fish corpora teach us

The available fish-AI datasets span very different definitions of an example. DeepFish contains roughly forty thousand in-situ frames from twenty habitats and supplies classification, localization/counting, and segmentation subsets; its authors emphasized natural background complexity and habitat-aware splits rather than only centered fish crops [2]. OzFish aggregates approximately eighty thousand fish crops and about forty-five thousand bounding-box annotations from baited remote underwater video, covering hundreds of species and linking each crop to family, genus, and species metadata [3]. Fishnet Open Images was built from fisheries electronic-monitoring imagery and contains 86,029 images across 34 object classes, deliberately retaining harsh weather, crew activity, and skewed class distributions [4]. A different FishNet benchmark organizes 94,532 images across 17,357 aquatic species using taxonomic hierarchy and includes detection annotations and functional traits [5]. A stock-estimation FishNet system reported a still larger operational collection of 300,000 hand-labeled images containing 1.2 million fish from 163 species, illustrating the scale possible when data collection is integrated into a field workflow [6].

FathomNet attacks a different bottleneck: standardization and aggregation across institutions. The peer-reviewed release introduced a global image database for ocean AI [7]; by 2026 the live database reported hundreds of thousands of images, more than a million bounding boxes, and a substantial segmentation-mask collection [8]. The dataset's contribution is not merely its size; it demonstrates that image URL, concept/taxon, localization geometry, depth, latitude, imaging type, and group annotations can coexist in a reusable schema. This is particularly important for rare species, because aggregating independently collected observations may be the only practical

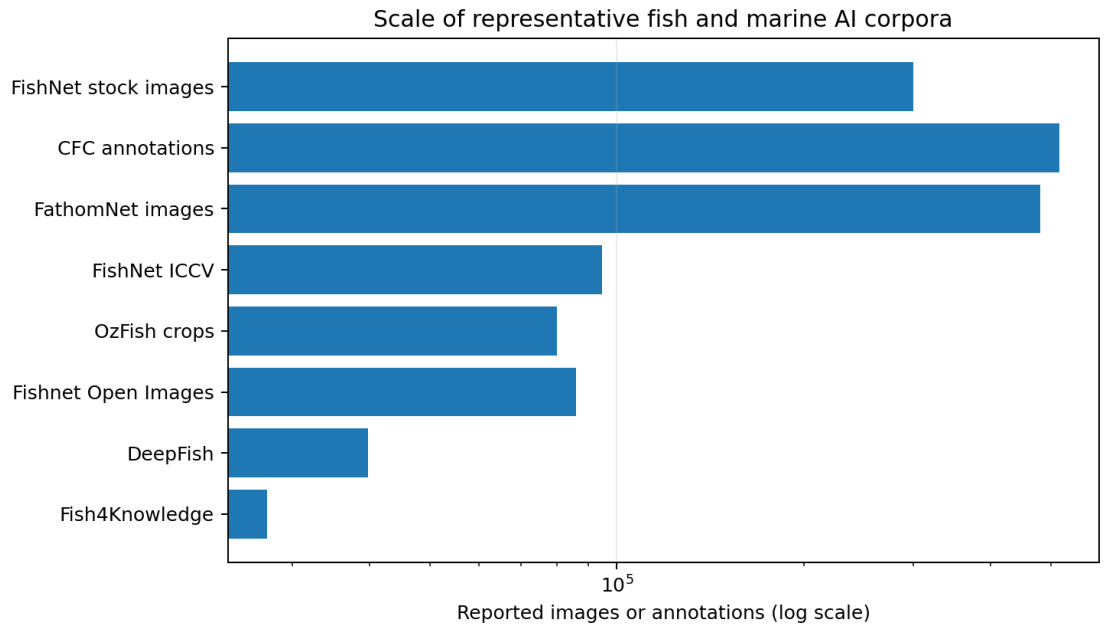


Figure 1: Scale of selected visual and sonar corpora. Counts are not directly comparable because some repositories report images whereas others report annotations. The figure is intended to show the order-of-magnitude range of current resources, not to rank dataset quality. Sources include DeepFish, Fishnet Open Images, OzFish, FishNet, FathomNet, and CFC [2–5, 7, 10].

path to diversity.

Sonar changes the corpus design problem. The Caltech Fish Counting Dataset contains more than half a million annotations in more than 1,500 sonar videos from seven cameras and was designed around multiple-object tracking, counting, and generalization to unseen test locations [10]. Scientific-data releases of DIDSON/ARIS observations likewise emphasize access to raw acoustic data as well as visualizations so future algorithms are not locked to one preprocessing choice [11]. A recent fish segmentation and tracking dataset uses semi-automatic masks propagated through video and reports iterative quality improvement, providing a useful pattern for annotation at scale: machine assistance is acceptable, but human verification and temporal consistency checks remain essential [12].

Acoustic and molecular corpora reveal still different constraints. FishSounds compiles global evidence and recordings of fish sound production and, in 2026, listed more than one thousand soniferous species; the limiting resource is often verified reference calls rather than raw audio volume [21, 22]. Reviews of passive acoustic monitoring stress that species, behavioral state, season, habitat, and recording configuration must accompany calls if the data are to serve as reliable natural acoustic tags [23–25]. For eDNA, multiple-marker strategies and the joint use of local and public reference databases are recommended to improve sensitivity and taxonomic reliability [27]; reference libraries should be voucher-linked and audited for taxonomic uncertainty [28, 29].

3 A three-tier architecture for fish-species corpora

The recommended architecture separates high-trust benchmark data from scalable training reservoirs. Tier A is a protected gold core: expert-verified examples from independent capture events with complete provenance, clear annotation policy, and fixed evaluation partitions. Tier B is the validated supervised set: larger quantities of boxes, masks, tracks, calls, or sequence

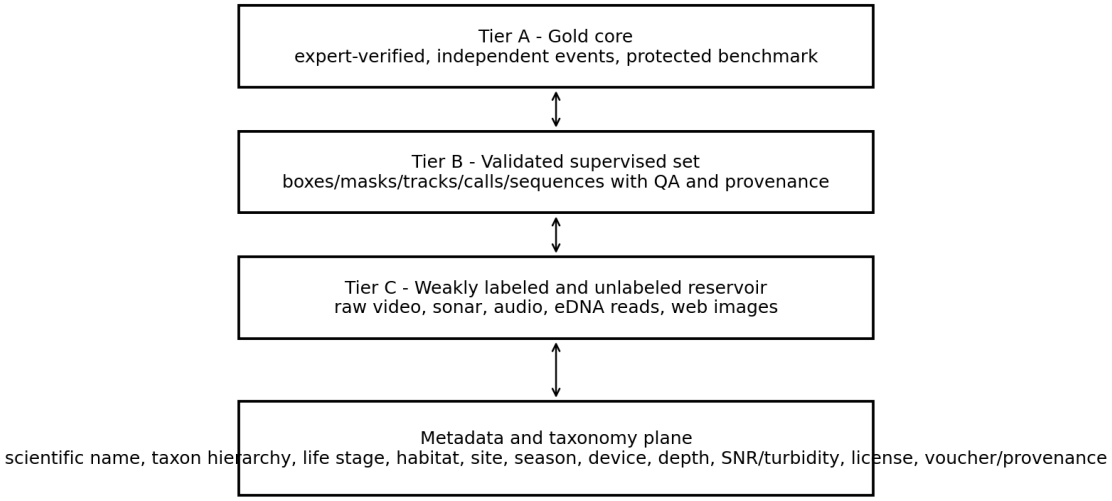


Figure 2: Recommended three-tier corpus architecture. Movement upward represents increasing verification and evaluation trust; movement downward represents increasing volume and lower annotation cost.

labels with quality-control sampling and adjudication. Tier C is a weakly labeled or unlabeled reservoir: raw video, sonar, hydrophone recordings, eDNA reads, opportunistic web imagery, or citizen-science observations that can support self-supervision, pretraining, active learning, or pseudo-labeling. The metadata plane spans all tiers and carries the taxonomic and environmental context needed to diagnose model failure.

A useful rule is to treat the independent biological or acquisition event, not the frame, as the atomic split unit. Let an event contain N frames whose label evidence is temporally autocorrelated. If the lag-k autocorrelation of a feature or outcome is ρ_k, an approximate effective sample size is

$$N_{\text{eff}} \approx \frac{N}{1 + 2\sum_{k=1}^{K} \rho_k}. \tag{1}$$

Ten thousand consecutive frames of one individual can therefore contribute far less independent information than one thousand frames sampled from hundreds of encounters. This is why sonar, underwater video, camera traps, and laboratory behavior experiments should split by clip, encounter, individual, tank, dive, station, cruise, or site. Camera-trap literature has repeatedly shown the danger of similar sequences occurring in both training and validation sets, a lesson that transfers directly to underwater video [14, 15]. The CFC benchmark's unseen-location tests provide a fish-specific example [10].

Tier A should not be consumed by routine model iteration. A practical arrangement is a development test set used sparingly and a final sealed benchmark used only for release decisions. For conservation-sensitive species, Tier A should include a "hard-known" set of diagnostically difficult examples and an "unknown/other" set so the system is evaluated on abstention rather than forced classification.

4 Taxonomy, labels, and the hierarchy problem

Fish recognition is inherently hierarchical. A specimen may be confidently identified to family or genus while species-level characters are occluded, absent in juveniles, or below image resolution.

FishNet explicitly organizes data by class, order, family, genus, and species [5]. A corpus should therefore retain the most specific defensible label rather than coercing every observation into a species class.

For each observation i, we recommend a taxonomic record

$$y_i = (c_i, o_i, f_i, g_i, s_i, q_i), \tag{2}$$

where c, o, f, g, s are taxonomic levels and q_i is label confidence or evidence quality. An observation can have $s_i =$ unknown while retaining a valid genus or family. This design supports hierarchical losses, open-set recognition, error analysis, and later relabeling when taxonomy changes. A targeted augmentation and hierarchical classification study for underwater fish recognition reported benefits from first reasoning at family level and then species level, consistent with the practical value of taxonomy-aware corpora [16].

Synonyms and taxonomic revisions should not silently rewrite historical labels. The manifest should store the original label, the accepted scientific name at corpus release time, the taxonomy authority/version, and an immutable internal taxon identifier. FishBase is suitable as a broad taxonomic reference, but image rights and species-level data provenance still need to be preserved separately [1].

The negative class deserves equal structure. "Background" should be decomposed into ecologically meaningful confusers: other fish, same-family fish, non-fish animals, vegetation, coral, substrate, bubbles, shadows, gear, humans, boats, acoustic artifacts, and empty scenes. For a species classifier, the hardest negative is usually a close taxonomic relative, not an empty frame. Consequently, the corpus budget should include deliberate collection of congener and confamily examples.

5 Species-first modality selection

No sensing modality is universally dominant. Optical imagery is information-rich when visibility and target size are adequate, but turbidity, depth, darkness, distance, schooling density, and camouflage can make image-only corpora fundamentally incomplete. Imaging sonar and scientific echosounders trade appearance detail for range and robustness to visibility. Passive acoustics is powerful for soniferous species and spawning choruses but cannot identify silent species by absence of sound. eDNA can reveal cryptic or rare presence but usually lacks individual count, pose, behavior, or instantaneous location. The best corpus therefore begins with observability analysis rather than model selection.

A practical species suitability vector is

$$z_s = [v_s, d_s, c_s, a_s, e_s, r_s], \tag{3}$$

where v is optical visibility/resolvability, d is density/occlusion risk, c is camouflage/fine-grained similarity, a is acoustic soniferousness or sonar suitability, e is value of eDNA/reference genetics, and r is rarity/conservation constraint. The corpus engineer should choose modalities that make the error modes complementary. For example, a salmon run may use sonar for passage detection and video/trap labels for species confirmation, whereas a cryptic reef fish may combine high-resolution images and eDNA presence confirmation.

6 Visual image and video corpus construction

6.1 Acquisition diversity

Natural underwater appearance varies with depth, solar angle, water color, turbidity, suspended particles, backscatter, camera white balance, compression, motion blur, lens port, strobe placement,

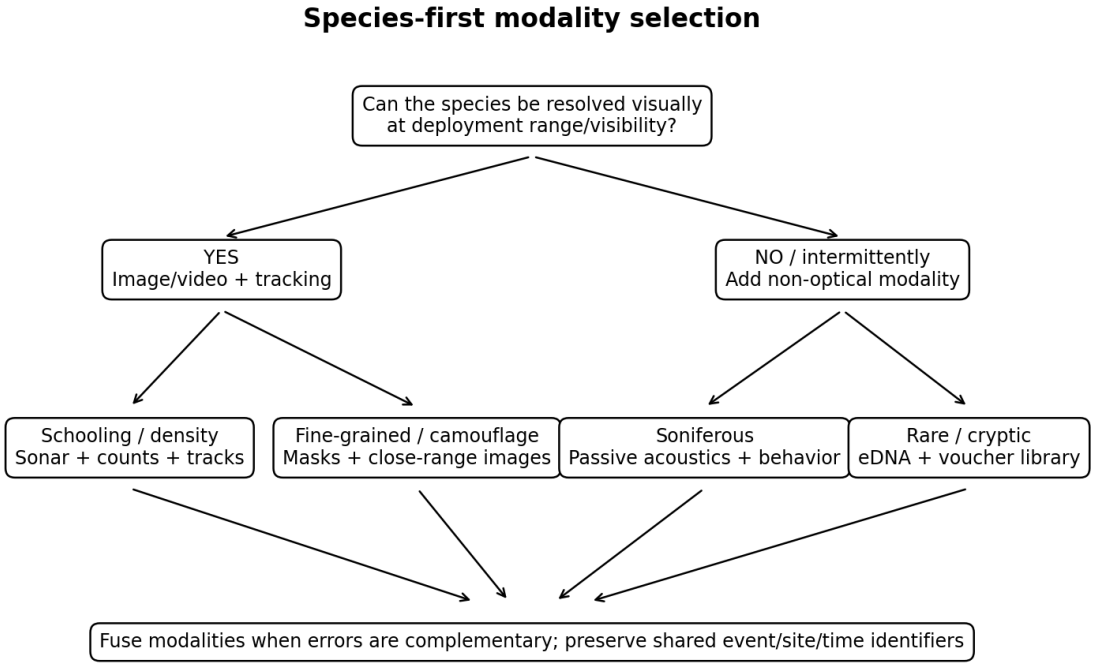


Figure 3: A species-first decision tree for selecting corpus modalities. Multimodal designs should retain shared event, site, and timestamp identifiers so observations can be fused without creating hidden duplicates.

focal length, viewing distance, and habitat. The corpus should model these variables explicitly. DeepFish was motivated by the limitations of centered, constrained fish crops and intentionally retained realistic habitat complexity [2]. OzFish similarly derives observations from BRUVS rather than only laboratory photographs [3]. A good acquisition plan therefore defines strata before collection: site, habitat, depth band, season, time of day, device, visibility class, range, orientation, life stage, group density, and behavior.

A useful coverage vector for species s is

$$\mathbf{c}_s = (c_{\text{site}}, c_{\text{season}}, c_{\text{stage}}, c_{\text{view}}, c_{\text{range}}, c_{\text{visibility}}, c_{\text{device}}, c_{\text{behavior}}). \tag{4}$$

Counts alone do not show coverage. A species with 20,000 images from one aquarium may have lower deployment coverage than a species with 2,000 images collected from twenty wild sites.

6.2 Annotation type should match the task

Image-level classification labels are cheapest but encourage shortcut learning from habitat. Bounding boxes support detection and reduce background dependence. Instance masks are preferred for camouflage, flatfish, dense overlap, length/shape estimation, or when future contour features are anticipated. Video should retain track IDs whenever counting, passage, behavior, or re-identification is in scope. The recent segmentation-and-tracking dataset demonstrates a practical semi-automatic workflow using model-generated masks, temporal propagation, human correction, and explicit measurement of identifier consistency [12].

For tiny objects, an annotation threshold must be defined. One defensible policy is to exclude targets below a resolution at which diagnostic structures cannot be labeled consistently, while

retaining them in a "present but unresolved" region label. Sanchez et al. use a size-based exclusion criterion for very small fish to reduce ambiguity [12]. The exact pixel threshold should be calibrated to camera resolution and task, not copied blindly.

6.3 Gold-standard annotation

Marine image annotation benefits from explicit preparation, annotation, quality assessment, and adjudication. RecoMIA recommends representative gold-standard sets and quantification of observer agreement [13]. We recommend double annotation for a stratified sample of every species and every difficult environmental stratum, not merely a global random sample. Disagreements should be categorized as taxonomic, localization, visibility, or policy ambiguities so guideline changes can be tracked.

7 Video independence, tracking, and counting corpora

For counting applications, a frame-level dataset is insufficient. The corpus must preserve continuity: track ID, first/last frame, entry/exit boundary, occlusion state, and whether an individual is genuinely new or a reappearance. CFC demonstrates the importance of tracking and counting annotations in sonar video and of testing generalization at unseen locations [10]. A species-specific passage corpus should similarly contain continuous clips rather than only positive snippets.

The recommended event unit is a *track episode*. An episode begins when an individual becomes observable and ends after it leaves the field of view or is unresolved beyond a predeclared gap. For each track, store species, confidence, trajectory, visibility fraction, direction, and optional length. Count metrics should be derived from boundary-crossing events rather than summing per-frame detections.

Data splits should be built from groups such as

$$G = (\text{site}, \text{camera}, \text{date}, \text{clip}, \text{encounter}), \tag{5}$$

with all frames from a group assigned to only one partition. Random frame splits can produce near-duplicate train and test images. A robust test additionally holds out entire cameras or sites.

8 Active acoustics: imaging sonar and scientific echosounders

Acoustic corpora require the raw measurement chain to be part of the data model. At minimum record sonar/echosounder model, frequency or band, pulse settings, range, gain/log compression, beam geometry, calibration status, mounting geometry, water depth, and preprocessing version. When licensing permits, release raw acoustic samples as well as derived images so researchers can revisit normalization and resampling. Scientific-data releases of DIDSON observations explicitly provide both raw and visualized forms for this reason [11].

For imaging sonar, labels should follow tracks through low signal-to-noise periods. Appearance descriptors are weak, so domain generalization is largely a function of site, range, bottom structure, current, sonar model, and processing. CFC's multiple cameras and unseen-location test are a strong template [10]. Multi-species acoustic-camera work also shows that a model can behave differently for different species and sonar models, motivating stratified performance reporting [17].

Scientific echosounder classification has a different unit: acoustic schools or pixels in multi-frequency echograms. Data-preparation choices such as resampling can remove class-relevant characteristics; contextual variables such as range can improve classification [18]. A corpus should therefore version the preprocessing pipeline and retain the original calibrated backscatter whenever practical.

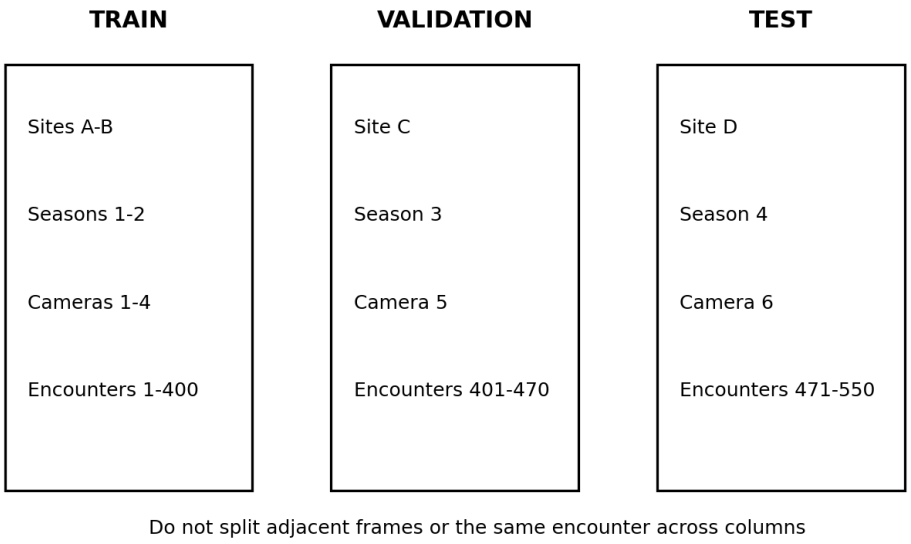


Figure 4: Group-wise splitting prevents leakage from adjacent frames, repeated encounters, cameras, and sites.

Species identification from echoes should be linked to independent evidence: trawl catches, optical imagery, known migration periods, or other biological sampling. Reviews of acoustic species classification emphasize uncertainty, class imbalance, and limited labeled data [19, 20]. The label should therefore encode evidence strength rather than pretending every school is perfectly identified.

9 Passive acoustic corpora for soniferous fishes

A passive-acoustic fish corpus is a library of verified sound events embedded in environmental context. The fundamental unit should not be an arbitrary fixed spectrogram window, because one call can be duplicated across overlapping windows and choruses can contain multiple callers. Store the original waveform, event start/end, sampling rate, hydrophone calibration, channel, site, depth, noise context, species, behavior, and verification method.

FishSounds demonstrates the importance of a global reference inventory and explicit links between taxa, recordings, and literature [21, 22]. Passive-acoustic reviews note that many fish families produce species-specific low-frequency sounds and that calls are associated with feeding, spawning, territorial, and other behaviors [23, 24]. A strong training corpus should therefore balance behavioral contexts rather than mixing them into one species label.

A reference call should ideally have synchronized video or direct observation, localization, or a controlled single-species recording. A recent dataset paired acoustic localization arrays with underwater video and extracted dozens of call features to establish species-specific reference sounds in the wild [26]. This paired-validation pattern is the acoustic analogue of voucher specimens in eDNA.

For each species, create three partitions: isolated high-confidence calls, natural mixed soundscapes with verified calls, and hard negatives containing abiotic transients and calls of related species. Evaluate event detection separately from species classification. When choruses are the deployment target, include chorus-level labels and time-of-day/seasonal context because individual-call performance does not guarantee chorus monitoring performance [25].

10 eDNA and genetic reference corpora

For cryptic, rare, nocturnal, or visually inaccessible fishes, an eDNA corpus may be the most reliable presence-detection resource. The critical training asset is not only environmental reads but a taxonomically correct reference library. Reviews recommend using multiple fish metabarcoding markers, such as 12S together with additional mitochondrial loci where justified, and combining curated local references with public databases [27]. Reference sequences should be connected to voucher specimens, sampling locality, marker, primers, sequencing platform, and taxonomic authority.

Let $R_{s,m}$ be the set of verified reference sequences for species s and marker m. A simple geographic coverage measure is

$$C_{s,m} = \frac{\text{represented biogeographic populations or regions}}{\text{target populations or regions}}. \tag{6}$$

A single sequence can identify a species in ideal conditions but cannot represent intraspecific diversity or reveal taxonomic conflicts. Barcode-library audits have uncovered uncertainties that become visible only after geographic expansion [28]. For management applications, the corpus should preserve ambiguity: a sequence may support genus-level assignment when species-level discrimination is not defensible.

Environmental samples need strict contamination metadata: field blank, extraction blank, PCR negative, positive control, replicate, filtration volume, storage, extraction batch, PCR batch, primer pair, read depth, and bioinformatic pipeline version. A recent guidance review emphasizes that primer choice, contamination, incomplete reference databases, and taxonomic assignment decisions can alter the reported species list; transparent limitations are therefore part of the corpus itself [29].

11 Multimodal corpus construction

A multimodal corpus is valuable only when the modalities are synchronized and provenance-preserving. The shared entity should be an event with a stable identifier:

$$e_j = (t_j, \ell_j, \text{site}_j, \text{platform}_j, \{x_j^{(m)}\}_{m=1}^{M}), \tag{7}$$

where the $x^{(m)}$ are video, sonar, audio, eDNA, environmental, or capture observations. If modalities are collected on different spatial or temporal scales, the relationship should be encoded as probabilistic or interval-based rather than falsely treated as exact.

Useful combinations include sonar plus video for migration counts, echosounder plus trawl/voucher labels for school classification, hydrophone plus localized video for soniferous species, and BRUVS plus eDNA for reef community monitoring. The objective is not to maximize the number of sensors; it is to create complementary evidence and robust labels. Multimodal data also supports teacher-student training in which a high-specificity modality provides labels for a lower-cost deployment modality.

12 Long-tail species, rare species, and open-set recognition

Fish datasets are naturally long-tailed: common species can produce millions of frames while rare species may have only a few encounters. Naive frame balancing can worsen the problem by oversampling nearly identical frames. Balance should operate on independent events. For species s with E_s independent events, a simple event-balanced sampling weight is

$$w_s \propto (E_s + \epsilon)^{-\alpha}, \qquad 0 \le \alpha \le 1. \tag{8}$$

This can be combined with taxonomy-aware sampling so batches contain hard confusers from the same family or genus.

For rare species, the recommended strategy is to maximize information per observation: expert verification, full metadata, masks if feasible, diagnostic-part visibility, and known confusers. Self-supervised pretraining on the unlabeled reservoir and hierarchical transfer from related taxa are usually preferable to aggressive synthetic duplication. Active learning can reduce annotation cost by selecting uncertain or diverse examples; fish-specific work has explored active learning for underwater species recognition [30].

Every operational classifier should include an abstain/unknown mechanism. Evaluation should report not only closed-set accuracy but also unknown detection, hierarchical accuracy, calibration, and false positive rate on ecologically plausible non-target species. For endangered species, a false positive can trigger expensive management actions, whereas a false negative can hide a conservation signal; the threshold should therefore be chosen from decision costs rather than a generic 0.5 probability.

13 Augmentation, synthetic data, and representation learning

Augmentation should mimic the deployment nuisance variables without changing species identity. Useful optical transforms include modest brightness/contrast variation, underwater color casts, blur, compression, particle backscatter, range-dependent resolution, and partial occlusion. Geometric flips or rotations should be used only when physically plausible for the camera geometry and task. For fine-grained species, strong color jitter can destroy diagnostic coloration and create label-preserving assumptions that are biologically false.

Targeted augmentation is better than uniform expansion of already dominant classes. Underwater fish recognition work has proposed augmenting classes selectively based on learning behavior and combining this with hierarchical classification [16]. More generally, augmentation should be driven by the corpus coverage vector: synthesize underrepresented visibility or viewpoint states, not arbitrary variants.

Synthetic compositing can help sonar counting or rare visual conditions but must be separated from the gold test set. A sonar counting study demonstrated the value of simulated augmentation under extremely limited labeled data [31]. The correct role of synthetic data is to supplement real diversity, never to replace field validation. Self-supervised learning can exploit abundant unlabeled sonar or video before fine-tuning; fish-counting work has shown this pattern can improve data efficiency [32].

14 Corpus sizing: engineering targets rather than universal constants

No publication establishes a universal minimum number of images per fish species. Required size depends on within-species appearance variance, number and similarity of confusers, sensor resolution, domain shift, and model pretraining. We therefore recommend using target ranges as planning aids and then stopping based on coverage and learning curves.

For a common visually resolvable species, an initial engineering target can be 500–2,000 independent positive encounters for classification, 5,000–20,000 annotated object instances for detection across those encounters, and 50–200 continuous videos for tracking/counting. Rare species may have only 50–200 verified encounters, in which case metadata quality, hierarchy, self-supervision, and open-set evaluation become more important than trying to meet an arbitrary count. For passive acoustics, hundreds of verified calls from multiple individuals/sites/behaviors

are a useful starting scale when biologically available. For eDNA reference libraries, multiple voucher-linked sequences across the target geographic range are preferred to a single canonical sequence.

These values are intentionally labeled *engineering targets*, not scientific standards. The stopping criterion should combine uncertainty in learning curves, coverage gaps, and marginal utility of new data. Define an adequacy score

$$A_s = \sum_{k=1}^{K} w_k \min(1, c_{s,k}/\tau_k), \tag{9}$$

where $c_{s,k}$ is achieved coverage for dimension k (sites, life stages, viewpoints, devices, seasons, confusers, visibility) and τ_k is the project's target. New acquisition should focus on the dimensions contributing most to $1 - A_s$ rather than collecting more of the easiest condition.

15 Data splitting, benchmark design, and leakage control

A benchmark should answer progressively harder questions. We recommend five evaluation tiers:

1. **In-domain**: new encounters from known sites/devices.
2. **Temporal**: new season or year at known sites.
3. **Device**: unseen camera/sonar/hydrophone configuration.
4. **Geographic**: unseen site, reef, river, farm, cruise, or seamount.
5. **Open-world**: target species mixed with unseen taxa and realistic non-fish negatives.

A single random split mostly measures Tier 1. For deployment claims, at least one Tier 3 or Tier 4 benchmark should be reported.

For video, the split key should include the parent source and encounter. For web imagery, near-duplicate and photographer/source clustering is needed because the same photograph can be reposted or cropped. For laboratory model fish, tank, trial, individual, and experiment are potential leakage units. For eDNA, sampling site and laboratory batch can leak contextual signatures. For passive acoustics, long recordings should not be cut into overlapping train/test windows.

The test corpus should be frozen with a versioned manifest and hashes. If taxonomic labels are later corrected, publish a new benchmark version and maintain a changelog rather than silently replacing files.

16 Quality control and annotation governance

The annotation handbook is part of the corpus. It should define species names and synonyms, life-stage rules, uncertain labels, minimum visible fraction, how to annotate groups/schools, occlusion, truncation, partial fish, dead specimens, reflections, and "too small to identify" cases. FathomNet's ingestion guidance illustrates the practical value of explicit localization columns and contextual metadata [9]. RecoMIA provides a general framework for observer quality and gold standards [13].

A recommended QA cycle is: (1) annotate a pilot batch; (2) calculate agreement and inspect disagreements; (3) revise guidelines; (4) re-annotate a fixed gold subset; (5) begin production annotation; (6) continuously audit stratified random samples and model-disagreement samples;

(7) adjudicate taxonomically difficult cases. Semi-automatic mask or track propagation should log the model/version used and the amount of human correction.

Quality statistics should be species-stratified. A global 95% box agreement can hide a 60% agreement for a cryptic target. For masks use IoU or boundary metrics; for boxes use IoU and category agreement; for tracks use ID continuity; for calls use event-boundary agreement and species agreement; for eDNA use replicate concordance and control status.

17 Metadata schema

The minimal cross-modal event manifest should include:

```
event_id, parent_recording_id, timestamp_start, timestamp_end,
scientific_name, taxon_id, family, genus, species,
label_rank, label_confidence, verifier_id, voucher_id,
life_stage, sex_if_known, behavior, count,
site_id, latitude, longitude, depth_m,
habitat, season, water_temperature, turbidity_or_visibility,
platform, sensor_model, sensor_settings, calibration_id,
file_uri, file_hash, license, source, annotation_version,
split_group, train_val_test
```

Modality-specific fields are then attached in side tables: image dimensions and boxes/masks/tracks; sonar frequency and range geometry; hydrophone calibration and sample rate; eDNA primer/marker and laboratory batches. The split-group field is especially important because it encodes the independence unit before downstream researchers create accidental frame leakage.

18 A universal decision procedure for any fish species

For a species not explicitly profiled later in this handbook, apply the following procedure.

1. **Define the decision**: presence/absence, species ID, count, biomass, size, behavior, health, individual re-ID, or community composition.
2. **Identify confusers**: same genus/family, same silhouette, co-occurring species, substrate and sensor artifacts.
3. **Score observability**: optical resolution, camouflage, density, behavior, depth/turbidity, soniferousness, sonar contrast, eDNA utility, rarity.
4. **Choose primary and secondary modalities**: prefer complementary error modes.
5. **Define the independent event**: encounter, track, school, call, water sample, specimen, trial, or dive.
6. **Design coverage strata**: site, season, stage, sex/color phase, viewpoint, range, device, visibility, behavior, density.
7. **Specify annotation**: image label, box, mask, track, call event, sequence label, marker sequence, and taxonomic confidence.
8. **Build Tier A first**: a small high-trust gold core including hard confusers and unknowns.
9. **Scale Tier B/C**: active learning, self-supervision, semi-automatic annotation, and public data aggregation.

10. **Split by group**: site/encounter/individual/source, not frame.
11. **Evaluate open-world**: unknown taxa, new sites/devices, and realistic background.
12. **Iterate acquisition from failures**: new data should close measured coverage gaps.

This procedure is the practical meaning of "best corpus by species": the protocol is generated from the species' observability and ecology rather than from its common name alone.

19 Species archetypes and how to read the protocol cards

The following 64 species cards are organized as worked examples. Each card names a preferred primary and secondary modality, annotation schema, split unit, hard negatives, sampling priorities, and evaluation suite. Recommendations are deliberately conservative: they favor independent events and environmental coverage over raw frame count. The cards should be adapted to local biology, legal constraints, and the exact deployment sensor. FishBase can supply up-to-date taxonomy and broad species information, but project-specific ecologists should validate life-stage, geographic, and diagnostic assumptions [1].

The archetypes are not mutually exclusive. Atlantic cod is demersal but also acoustically surveyed; salmon are migratory but can be visually identified; whale sharks are large pelagics and elasmobranchs. The card assigns the archetype that most strongly influences corpus engineering for the example task.

20 Reference landscape of fish and marine corpora

Resource	Approx. scale	Main modality/task	Corpus-construction lesson
DeepFish [2]	39,766 frames	In-situ RGB; classification; counting; localization; segmentation	Natural habitat and background variability are part of the task; provide task-specific labels and habitat-aware splits.
OzFish [3]	∼80k crops, ∼45k boxes, 507 species	BRUVS RGB; species and detection	Preserve taxonomic hierarchy and source video; combine crop convenience with scene context.
Fishnet Open Images [4]	86,029 images, 34 classes	Fisheries electronic monitoring	Keep operational clutter, crew, weather and skewed class distributions; benchmark realistic monitoring conditions.
FishNet ICCV [5]	94,532 images, 17,357 aquatic species	Classification; detection; traits	Taxonomy-aware organization enables hierarchical learning and broad species coverage, but per-species tail remains sparse.
FishNet stock estimation [6]	300k images; 1.2M fish; 163 species	Detection, segmentation, species, length	Integrated field workflow plus quality control can reach operational scale and support multiple outputs.
FathomNet [7, 8]	Hundreds of thousands of images; >1M boxes	Marine imagery repository	Standardized concepts, localizations and provenance allow distributed aggregation and rare-taxon reuse.
Caltech Fish Counting [10]	>0.5M annotations; >1,500 videos	Imaging sonar; tracking and counting	Evaluate at unseen locations and preserve tracks; appearance cues are weak in sonar so site generalization matters.
FishSounds [21, 22]	>1,000 soniferous species; reference recordings	Passive acoustics	Verified reference sounds and behavior/context are the bottleneck; raw hours without source identity are insufficient.
DeepFish tracking 2026 [12]	Video masks and tracks	Segmentation and MOT	Machine-assisted propagation plus iterative human correction can scale annotation while retaining measurable quality.

20.1 1. Atlantic herring (*Clupea harengus*)

Archetype and observability. Atlantic herring is treated here as a *pelagic school* corpus problem in temperate marine pelagic. The operational behavior most relevant to data collection is dense schooling, rapid heading changes. A typical recognition cue is silvery laterally compressed body, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are Atlantic mackerel, sardines, sprat. Use repeated independent schools across seasons and light levels; include mixed-species schools and distant silhouettes.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.2 2. European sardine (*Sardina pilchardus*)

Archetype and observability. European sardine is treated here as a *pelagic school* corpus problem in coastal marine pelagic. The operational behavior most relevant to data collection is schooling, strong density variation. A typical recognition cue is small reflective clupeid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are anchovy, sprat, juvenile herring. Emphasize scale variation and schooling density; many labels should be school-level counts as well as individual boxes.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.3 3. European anchovy (*Engraulis encrasicolus*)

Archetype and observability. European anchovy is treated here as a *pelagic school* corpus problem in coastal pelagic and estuarine. The operational behavior most relevant to data collection is schooling, small target size. A typical recognition cue is slender small clupeiform, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are sardine, sprat, juvenile herring. Collect close-range taxonomic confirmation because low-resolution video frequently collapses anchovy and sardine appearance.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.4 4. Atlantic mackerel (*Scomber scombrus*)

Archetype and observability. Atlantic mackerel is treated here as a *pelagic school* corpus problem in temperate pelagic. The operational behavior most relevant to data collection is fast schools, strong orientation effects. A typical recognition cue is fusiform body with dorsal banding, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are horse mackerel, juvenile tuna, bonito. Ensure lateral, oblique and tail-on views; do not let distinctive lateral patterns dominate every training example.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.5 5. Capelin (*Mallotus villosus*)

Archetype and observability. Capelin is treated here as a *pelagic school* corpus problem in cold marine pelagic. The operational behavior most relevant to data collection is large seasonal aggregations. A typical recognition cue is small slender smelt-like body, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are sand lance, smelts, juvenile herring. Stratify by spawning/non-spawning seasons and size; sonar is often more scalable than optics for large aggregations.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.6 6. Walleye pollock (*Gadus chalcogrammus*)

Archetype and observability. Walleye pollock is treated here as a *pelagic school* corpus problem in subarctic marine, semi-pelagic. The operational behavior most relevant to data collection is schools and midwater layers. A typical recognition cue is gadid morphology, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are Pacific cod, hake, juvenile cod-like fishes. Combine trawl/voucher confirmation with acoustic layers; isolate cruise/site splits to avoid survey-specific signatures.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.7 7. Pacific sardine (*Sardinops sagax*)

Archetype and observability. Pacific sardine is treated here as a *pelagic school* corpus problem in eastern Pacific pelagic. The operational behavior most relevant to data collection is large mobile schools. A typical recognition cue is small silver clupeid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are anchovy, herring, jack mackerel juveniles. Include oceanographic regimes and multiple camera/sonar systems; evaluate on unseen cruises.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.8 8. Atlantic menhaden (*Brevoortia tyrannus*)

Archetype and observability. Atlantic menhaden is treated here as a *pelagic school* corpus problem in western Atlantic coastal pelagic. The operational behavior most relevant to data collection is dense near-surface schools. A typical recognition cue is deep-bodied clupeid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are herring, shad, sardines. Drone and near-surface imagery can complement sonar; include glare and wave-caustic negatives.

Recommended modalities. The preferred primary modality is video plus active acoustics/sonar. A complementary secondary source is eDNA and trawl/voucher confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. sample independent schools across density, depth, day/night, turbidity, range and orientation; retain empty-water intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store school polygons or density regions, individual boxes when resolvable, track IDs, school-size/count intervals, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include mixed schools, wave caustics, bubbles, plankton layers, similar small pelagics. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use cruise/site/season held-out; never random adjacent-frame splitting. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species AP/F1, count error by density, track recall, calibration, unseen-cruise performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.9 9. Atlantic bluefin tuna (*Thunnus thynnus*)

Archetype and observability. Atlantic bluefin tuna is treated here as a *large pelagic* corpus problem in open-ocean pelagic. The operational behavior most relevant to data collection is fast, migratory, often sparse. A typical recognition cue is large tuna fusiform profile, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are yellowfin tuna, bigeye tuna, albacore. Prioritize long-range size-normalized views, fin geometry, and multi-camera context; split by expedition and individual encounter.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.10 10. Yellowfin tuna (*Thunnus albacares*)

Archetype and observability. Yellowfin tuna is treated here as a *large pelagic* corpus problem in tropical and subtropical pelagic. The operational behavior most relevant to data collection is fast schooling and free-swimming. A typical recognition cue is tuna with distinctive fin proportions, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are bigeye tuna, bluefin tuna, albacore. Include juveniles where fin cues are weaker and hard negatives from other Thunnus species.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.11 11. Skipjack tuna (*Katsuwonus pelamis*)

Archetype and observability. Skipjack tuna is treated here as a *large pelagic* corpus problem in tropical pelagic. The operational behavior most relevant to data collection is fast schools near surface. A typical recognition cue is compact tuna with ventral striping, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are small yellowfin, bonito, little tunny. Use high shutter-speed and motion-blur examples; include partial bodies in purse-seine and onboard monitoring footage.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.12 12. Mahi-mahi (*Coryphaena hippurus*)

Archetype and observability. Mahi-mahi is treated here as a *large pelagic* corpus problem in warm offshore pelagic. The operational behavior most relevant to data collection is fast, surface-associated. A typical recognition cue is elongate body, pronounced head profile, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are wahoo, juvenile billfish, pompano dolphinfish. Capture color-state variability because stress and illumination alter apparent colors substantially.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.13 13. Swordfish (*Xiphias gladius*)

Archetype and observability. Swordfish is treated here as a *large pelagic* corpus problem in oceanic pelagic, deep vertical migrations. The operational behavior most relevant to data collection is solitary, fast, long-billed. A typical recognition cue is long rostrum and crescent tail, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are sailfish, marlins. Long-range imagery should label partial rostrum/body cases separately; include day/night and deck-camera domains.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.14 14. Sailfish (*Istiophorus platypterus*)

Archetype and observability. Sailfish is treated here as a *large pelagic* corpus problem in warm pelagic. The operational behavior most relevant to data collection is fast, predatory, surface-associated. A typical recognition cue is large sail-like dorsal fin and bill, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are marlin, swordfish. Annotate fin-folded states and partial surface breaches; the corpus should not rely only on iconic fully displayed sails.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.15 15. Ocean sunfish (*Mola mola*)

Archetype and observability. Ocean sunfish is treated here as a *large pelagic* corpus problem in oceanic epipelagic to mesopelagic. The operational behavior most relevant to data collection is slow cruising, basking. A typical recognition cue is very deep truncated disk-like body, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other molids, large rays at distance. Include multiple life stages and distant silhouettes; object scale can vary by orders of magnitude.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.16 16. Whale shark (*Rhincodon typus*)

Archetype and observability. Whale shark is treated here as a *large pelagic* corpus problem in tropical oceanic/coastal. The operational behavior most relevant to data collection is large slow filter feeder. A typical recognition cue is large shark with spotted pattern, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other large sharks, manta rays. Treat individual re-identification as a second task using flank spot patterns; maintain encounter-level identity splits.

Recommended modalities. The preferred primary modality is long-range video/stereo imagery with encounter metadata. A complementary secondary source is sonar, tagging context, opportunistic citizen-science imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. collect near/far views, partial bodies, different camera platforms and ocean states; avoid iconic-photo bias. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, body orientation, visibility fraction, life stage, size proxy, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other large pelagics, partial silhouettes, glare, surface foam, vessels and gear. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use encounter/individual/expedition held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy under distance/occlusion, open-set rejection, individual re-ID where applicable. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.17 17. Atlantic cod (*Gadus morhua*)

Archetype and observability. Atlantic cod is treated here as a *demersal* corpus problem in temperate/subarctic demersal. The operational behavior most relevant to data collection is schools and benthic association. A typical recognition cue is gadid with chin barbel, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are haddock, pollock, hake. Mix optics with broadband acoustics when possible; include age/size classes and benthic clutter.

Recommended modalities. The preferred primary modality is underwater video plus echosounder/broadband acoustic context. A complementary secondary source is trawl/deck voucher images and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover seabed types, depth bands, low-light, turbidity and near-bottom boundary conditions. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, substrate, range/depth, orientation, length, life stage, school context. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include closely related demersal fishes, benthic debris, partial fish near seabed. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use survey cruise/site held-out with gear-specific domains identified. A random frame split is not acceptable for a deployment claim. The evaluation suite should include fine-grained species confusion matrix, detection near bottom, acoustic/visual cross-domain transfer. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.18 18. Haddock (*Melanogrammus aeglefinus*)

Archetype and observability. Haddock is treated here as a *demersal* corpus problem in North Atlantic demersal. The operational behavior most relevant to data collection is schooling demersal. A typical recognition cue is gadid with dark lateral mark, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are Atlantic cod, whiting, pollock. Hard-negative curation against cod is more valuable than generic background expansion.

Recommended modalities. The preferred primary modality is underwater video plus echosounder/broadband acoustic context. A complementary secondary source is trawl/deck voucher images and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover seabed types, depth bands, low-light, turbidity and near-bottom boundary conditions. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, substrate, range/depth, orientation, length, life stage, school context. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include closely related demersal fishes, benthic debris, partial fish near seabed. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use survey cruise/site held-out with gear-specific domains identified. A random frame split is not acceptable for a deployment claim. The evaluation suite should include fine-grained species confusion matrix, detection near bottom, acoustic/visual cross-domain transfer. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.19 19. European hake (*Merluccius merluccius*)

Archetype and observability. European hake is treated here as a *demersal* corpus problem in continental shelf/slope. The operational behavior most relevant to data collection is demersal, nocturnal vertical movement. A typical recognition cue is elongate gadiform, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are pollock, whiting, juvenile cod. Collect low-light and trawl-deck images as separate domains; evaluate domain transfer explicitly.

Recommended modalities. The preferred primary modality is underwater video plus echosounder/broadband acoustic context. A complementary secondary source is trawl/deck voucher images and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover seabed types, depth bands, low-light, turbidity and near-bottom boundary conditions. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, substrate, range/depth, orientation, length, life stage, school context. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include closely related demersal fishes, benthic debris, partial fish near seabed. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use survey cruise/site held-out with gear-specific domains identified. A random frame split is not acceptable for a deployment claim. The evaluation suite should include fine-grained species confusion matrix, detection near bottom, acoustic/visual cross-domain transfer. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.20 20. Pacific halibut (*Hippoglossus stenolepis*)

Archetype and observability. Pacific halibut is treated here as a *flatfish* corpus problem in North Pacific demersal. The operational behavior most relevant to data collection is benthic, often partially buried. A typical recognition cue is large right-eyed flatfish, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are arrowtooth flounder, other flatfishes. Top-down and oblique benthic views matter more than side-view museum images; include substrate-matched hard negatives.

Recommended modalities. The preferred primary modality is top/oblique benthic video with instance masks. A complementary secondary source is specimen imagery and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. oversample low-contrast and partially buried states rather than clean specimens. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store full masks, eye-side/visible-side, burial fraction, substrate type, orientation and size. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other flatfishes, rays, leaves, rocks, sand ripples and partially buried shapes. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/substrate/encounter held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, species accuracy by burial fraction, open-set rate on unknown flatfish. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.21 21. European plaice (*Pleuronectes platessa*)

Archetype and observability. European plaice is treated here as a *flatfish* corpus problem in European shelf demersal. The operational behavior most relevant to data collection is benthic camouflage. A typical recognition cue is flatfish with spotted dorsal surface, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are flounder, dab, sole. Segmentation masks and substrate diversity are preferred to tight crops; keep color augmentation conservative.

Recommended modalities. The preferred primary modality is top/oblique benthic video with instance masks. A complementary secondary source is specimen imagery and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. oversample low-contrast and partially buried states rather than clean specimens. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store full masks, eye-side/visible-side, burial fraction, substrate type, orientation and size. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other flatfishes, rays, leaves, rocks, sand ripples and partially buried shapes. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/substrate/encounter held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, species accuracy by burial fraction, open-set rate on unknown flatfish. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.22 22. Common sole (*Solea solea*)

Archetype and observability. Common sole is treated here as a *flatfish* corpus problem in coastal demersal. The operational behavior most relevant to data collection is benthic and cryptic. A typical recognition cue is elongate oval flatfish, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other soles and small flatfish. Collect burial depth variation and night footage; label "uncertain flatfish" when diagnostic characters are unresolved.

Recommended modalities. The preferred primary modality is top/oblique benthic video with instance masks. A complementary secondary source is specimen imagery and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. oversample low-contrast and partially buried states rather than clean specimens. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store full masks, eye-side/visible-side, burial fraction, substrate type, orientation and size. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other flatfishes, rays, leaves, rocks, sand ripples and partially buried shapes. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/substrate/encounter held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, species accuracy by burial fraction, open-set rate on unknown flatfish. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.23 23. Monkfish (*Lophius piscatorius*)

Archetype and observability. Monkfish is treated here as a *camouflage* corpus problem in shelf and slope benthic. The operational behavior most relevant to data collection is sit-and-wait predator. A typical recognition cue is broad flattened head, cryptic coloration, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are scorpionfish, rays, substrate objects. Use masks and contour labels; negative mining from rocks/sponges is essential.

Recommended modalities. The preferred primary modality is high-resolution video/stills with segmentation masks and background negatives. A complementary secondary source is macro specimen photos, habitat context, eDNA for presence confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. actively mine false positives from natural substrate; include low contrast and partial visibility. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks, visibility fraction, occlusion, substrate/habitat, uncertainty flag, diagnostic-part visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include substrate structures, coral, algae, sponges, debris and related cryptic taxa. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/habitat held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, recall at low visibility, false positives per hour, uncertainty calibration. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.24 24. Orange roughy (*Hoplostethus atlanticus*)

Archetype and observability. Orange roughy is treated here as a *deep sea* corpus problem in deep slope/seamount. The operational behavior most relevant to data collection is slow aggregations at depth. A typical recognition cue is deep-bodied reddish fish, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are oreo dories, alfonsino, other deep fishes. ROV illumination and depth metadata are critical; reserve entire seamounts or cruises for test.

Recommended modalities. The preferred primary modality is ROV/AUV imagery with synchronized depth, vehicle, lighting and navigation metadata. A complementary secondary source is specimen/voucher photographs, eDNA, sonar where available. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. preserve raw low-light data; cover multiple vehicles/lighting geometries and rare encounter contexts. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, taxonomic confidence, depth, range, lighting state, life stage, specimen link. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other deep-sea fishes, shadows, marine snow, vehicle artifacts. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use dive/expedition/seamount held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include open-set metrics, taxonomic hierarchical accuracy, calibration and expedition-held-out recall. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.25 25. Atlantic salmon (*Salmo salar*)

Archetype and observability. Atlantic salmon is treated here as a *migratory* corpus problem in rivers and North Atlantic. The operational behavior most relevant to data collection is anadromous migration, rapid river passage. A typical recognition cue is salmonid body, life-stage changes, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are brown trout, sea trout, other salmonids. Sonar/video corpora should preserve direction, passage event and camera site; separate smolt, adult and spawning phenotypes.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.26 26. Chinook salmon (*Oncorhynchus tshawytscha*)

Archetype and observability. Chinook salmon is treated here as a *migratory* corpus problem in North Pacific and rivers. The operational behavior most relevant to data collection is anadromous, large adult migration. A typical recognition cue is large Pacific salmon, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are coho salmon, chum salmon, steelhead. Train with site-specific river optics and sonar but test on unseen rivers; record run timing and life stage.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.27 27. Sockeye salmon (*Oncorhynchus nerka*)

Archetype and observability. Sockeye salmon is treated here as a *migratory* corpus problem in North Pacific and lakes/rivers. The operational behavior most relevant to data collection is anadromous schooling migration. A typical recognition cue is strong spawning color change, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are coho, chum, kokanee. Corpus must include ocean, river, and spawning coloration; a single seasonal appearance is insufficient.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.28 28. Rainbow trout (*Oncorhynchus mykiss*)

Archetype and observability. Rainbow trout is treated here as a *migratory* corpus problem in freshwater and anadromous forms. The operational behavior most relevant to data collection is variable resident/anadromous behavior. A typical recognition cue is trout morphology with spotting, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are cutthroat trout, juvenile salmon, brown trout. Include hatchery/wild domains and juvenile/adult sizes; avoid waterbody leakage.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.29 29. Brown trout (*Salmo trutta*)

Archetype and observability. Brown trout is treated here as a *migratory* corpus problem in freshwater and sea-run forms. The operational behavior most relevant to data collection is resident and migratory. A typical recognition cue is brown/gold trout with spotting, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are Atlantic salmon parr, rainbow trout. Include parr, adult river, and sea-trout phenotypes; hierarchical salmonid labels help ambiguous juveniles.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.30 30. Arctic char (*Salvelinus alpinus*)

Archetype and observability. Arctic char is treated here as a *migratory* corpus problem in cold freshwater and Arctic coastal. The operational behavior most relevant to data collection is resident/anadromous. A typical recognition cue is char morphology, seasonal color, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are brook trout, Dolly Varden, salmonids. Geographic population variation is substantial; site-held-out evaluation is mandatory.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.31 31. European eel (*Anguilla anguilla*)

Archetype and observability. European eel is treated here as a *eel* corpus problem in freshwater, estuarine and marine migration. The operational behavior most relevant to data collection is elongate flexible body, nocturnal. A typical recognition cue is elongate anguilliform silhouette, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are American eel, lamprey, ropes/debris in sonar. Use centerlines/instance masks and flexion states; sonar corpus should include bottom clutter and low-SNR partial tracks.

Recommended modalities. The preferred primary modality is imaging sonar plus low-light video where possible. A complementary secondary source is trap/voucher labels and eDNA. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture full body-flexion distribution, bottom-following tracks and partial low-SNR detections. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store centerline or mask, head/tail if resolvable, track ID, flexion state, direction, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include lampreys, snakes, ropes, vegetation, turbulence traces. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/night/season held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include track continuity, centerline accuracy, species recall and false tracks per hour. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.32 32. American shad (*Alosa sapidissima*)

Archetype and observability. American shad is treated here as a *migratory* corpus problem in western Atlantic rivers/coast. The operational behavior most relevant to data collection is anadromous schooling migration. A typical recognition cue is deep-bodied clupeid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are river herring, alewife, menhaden. For passage monitoring, species labels require synchronized video/trap confirmation because sonar appearance is weakly diagnostic.

Recommended modalities. The preferred primary modality is fixed video and/or imaging sonar at migration passages. A complementary secondary source is trap counts, PIT/tag data, eDNA and expert visual confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. cover flow, turbidity, range, crowding and migration timing; retain continuous negative intervals. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store track IDs, direction, passage event, species, life stage, body length, occlusion and confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other migratory fishes, debris, bubbles, shadows, reverse-direction tracks. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use river/site/run-year held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include passage count error, track recall, species accuracy, unseen-river generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.33 33. Nile tilapia (*Oreochromis niloticus*)

Archetype and observability. Nile tilapia is treated here as a *aquaculture* corpus problem in freshwater tropical, aquaculture. The operational behavior most relevant to data collection is dense tanks/cages, social schooling. A typical recognition cue is deep-bodied cichlid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other tilapia species, hybrids. Capture crowding, net occlusion, feeding turbulence and size classes; instance segmentation plus tracking is preferred.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.34 34. Common carp (*Cyprinus carpio*)

Archetype and observability. Common carp is treated here as a *aquaculture* corpus problem in freshwater rivers/lakes/ponds. The operational behavior most relevant to data collection is benthic feeding, schooling. A typical recognition cue is large cyprinid with variable color, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are crucian carp, grass carp, koi variants. Include wild, farm and ornamental color morphs as explicit domains or sublabels.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.35 35. Grass carp (*Ctenopharyngodon idella*)

Archetype and observability. Grass carp is treated here as a *aquaculture* corpus problem in freshwater, cultured and introduced. The operational behavior most relevant to data collection is large schooling herbivore. A typical recognition cue is elongate cyprinid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are common carp, black carp, silver carp. Length estimation needs calibrated views; hard negatives should include other Asian carps.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.36 36. Silver carp (*Hypophthalmichthys molitrix*)

Archetype and observability. Silver carp is treated here as a *aquaculture* corpus problem in large rivers/reservoirs, aquaculture. The operational behavior most relevant to data collection is schooling, surface-jumping response. A typical recognition cue is deep-bodied Asian carp, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are bighead carp, juvenile grass carp. Collect airborne/partial-jump frames if boat monitoring is intended; include turbidity gradients.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.37 37. Channel catfish (*Ictalurus punctatus*)

Archetype and observability. Channel catfish is treated here as a *aquaculture* corpus problem in freshwater rivers/ponds, farmed. The operational behavior most relevant to data collection is benthic, often turbid water. A typical recognition cue is catfish with barbels, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are blue catfish, flathead catfish. Optical visibility may be poor; pair near-field images with sonar or passive acoustic events in suitable contexts.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.38 38. Striped catfish (*Pangasianodon hypophthalmus*)

Archetype and observability. Striped catfish is treated here as a *aquaculture* corpus problem in Southeast Asian rivers and aquaculture. The operational behavior most relevant to data collection is dense cultured schools. A typical recognition cue is pangasiid catfish, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other pangasiids, catfish juveniles. Build camera-domain diversity across farms because tank color and water opacity can dominate models.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.39 39. Rohu (*Labeo rohita*)

Archetype and observability. Rohu is treated here as a *aquaculture* corpus problem in South Asian freshwater aquaculture. The operational behavior most relevant to data collection is schooling column feeder. A typical recognition cue is large carp-like cyprinid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are catla, mrigal, common carp. Use multi-species pond scenes rather than single-fish trays; annotate co-cultured species as hard negatives.

Recommended modalities. The preferred primary modality is fixed multi-camera video with calibrated geometry. A complementary secondary source is sonar, biomass sensors, individual tags and water-quality telemetry. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. capture feeding/non-feeding, high/low density, day/night and multiple farms; prevent tank-ID shortcut learning. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks/boxes, track IDs, length/weight proxy, health/behavior attributes, crowding and visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include conspecific overlap, nets, feed pellets, bubbles, reflections and co-cultured species. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use tank/cage/farm/day held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include count/biomass error, track fragmentation, size error, cross-farm generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.40 40. Largemouth bass (*Micropterus salmoides*)

Archetype and observability. Largemouth bass is treated here as a *freshwater* corpus problem in lakes/reservoirs, aquaculture/game fish. The operational behavior most relevant to data collection is solitary/territorial predator. A typical recognition cue is centrarchid body, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are smallmouth bass, spotted bass. Capture juvenile and adult body-shape changes, vegetation occlusion and recreational-hold images as separate domains.

Recommended modalities. The preferred primary modality is video/stills in natural habitat with site metadata. A complementary secondary source is eDNA, electrofishing/net survey labels, sonar in turbid systems. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. include vegetation, turbidity, seasonal color and multiple waterbodies. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, size, life stage, habitat structure, turbidity, species confidence. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include related freshwater species, vegetation, reflections and substrate. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use waterbody/site held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species accuracy, detection recall, open-set unknown handling, unseen-waterbody results. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.41 41. Ocellaris clownfish (*Amphiprion ocellaris*)

Archetype and observability. Ocellaris clownfish is treated here as a *reef fine* corpus problem in tropical reef/anemone association. The operational behavior most relevant to data collection is site-attached, pair/group. A typical recognition cue is orange-white-black banding, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are percula clownfish, other Amphiprion. Include host anemone variability and juvenile patterning; use genus-level fallback when diagnostic characters are hidden.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.42 42. Blue tang (*Paracanthurus hepatus*)

Archetype and observability. Blue tang is treated here as a *reef fine* corpus problem in Indo-Pacific coral reef. The operational behavior most relevant to data collection is active reef swimmer. A typical recognition cue is blue body with dark pattern, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are surgeonfishes, juvenile color variants. Include juvenile/adult color forms and blue-light/white-balance variation.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.43 43. Yellow tang (*Zebrasoma flavescens*)

Archetype and observability. Yellow tang is treated here as a *reef fine* corpus problem in Pacific reef. The operational behavior most relevant to data collection is grazing, active. A typical recognition cue is bright yellow disk-like surgeonfish, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other Zebrasoma, similarly colored butterflyfishes. Avoid shortcut learning from aquarium blue backgrounds; include wild reef scenes and night color changes.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.44 44. Queen angelfish (*Holacanthus ciliaris*)

Archetype and observability. Queen angelfish is treated here as a *reef fine* corpus problem in western Atlantic coral reef. The operational behavior most relevant to data collection is territorial reef fish. A typical recognition cue is deep body, vivid pattern, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are blue angelfish, juvenile angelfishes. Life-stage labels are essential because juvenile patterns differ strongly from adults.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.45 45. Four-eye butterflyfish (*Chaetodon capistratus*)

Archetype and observability. Four-eye butterflyfish is treated here as a *reef fine* corpus problem in western Atlantic reef. The operational behavior most relevant to data collection is pairing, reef-associated. A typical recognition cue is disk-like body with eye band/ocellus, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other Chaetodon species. Fine-grained family confusers require high-resolution flank views plus occluded and oblique views.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.46 46. Stoplight parrotfish (*Sparisoma viride*)

Archetype and observability. Stoplight parrotfish is treated here as a *reef fine* corpus problem in Caribbean reef. The operational behavior most relevant to data collection is grazing, phase/color changes. A typical recognition cue is parrotfish beak, phase-dependent colors, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other parrotfish, terminal/initial phases. Treat color phase as an attribute, not a separate species; balance phases to prevent color shortcut learning.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.47 47. Nassau grouper (*Epinephelus striatus*)

Archetype and observability. Nassau grouper is treated here as a *reef fine* corpus problem in Caribbean reef. The operational behavior most relevant to data collection is solitary/aggregating spawner. A typical recognition cue is robust serranid with bars, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other groupers. Include spawning aggregation density and color-state changes; location-held-out tests are important for conservation monitoring.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.48 48. Red snapper (*Lutjanus campechanus*)

Archetype and observability. Red snapper is treated here as a *reef fine* corpus problem in western Atlantic/Gulf reef and shelf. The operational behavior most relevant to data collection is schooling around structure. A typical recognition cue is red lutjanid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are vermilion snapper, silk snapper, other Lutjanus. Collect depth-related color attenuation and deck/underwater domains separately; include morphologically similar snappers.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.49 49. Red lionfish (*Pterois volitans*)

Archetype and observability. Red lionfish is treated here as a *camouflage* corpus problem in tropical reef, invasive Atlantic. The operational behavior most relevant to data collection is slow hovering predator. A typical recognition cue is striped body with long spines, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are scorpionfish, coral branches. Segmentation masks should include fin rays; hard negatives from branching coral are critical.

Recommended modalities. The preferred primary modality is high-resolution video/stills with segmentation masks and background negatives. A complementary secondary source is macro specimen photos, habitat context, eDNA for presence confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. actively mine false positives from natural substrate; include low contrast and partial visibility. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks, visibility fraction, occlusion, substrate/habitat, uncertainty flag, diagnostic-part visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include substrate structures, coral, algae, sponges, debris and related cryptic taxa. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/habitat held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, recall at low visibility, false positives per hour, uncertainty calibration. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.50 50. Reef stonefish (*Synanceia verrucosa*)

Archetype and observability. Reef stonefish is treated here as a *camouflage* corpus problem in Indo-Pacific reef/benthic. The operational behavior most relevant to data collection is cryptic, sedentary. A typical recognition cue is irregular rock-like body, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are rocks, scorpionfish, coral rubble. Background-only corpus is as important as positives; use expert review and uncertainty labels for partially visible individuals.

Recommended modalities. The preferred primary modality is high-resolution video/stills with segmentation masks and background negatives. A complementary secondary source is macro specimen photos, habitat context, eDNA for presence confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. actively mine false positives from natural substrate; include low contrast and partial visibility. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks, visibility fraction, occlusion, substrate/habitat, uncertainty flag, diagnostic-part visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include substrate structures, coral, algae, sponges, debris and related cryptic taxa. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/habitat held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, recall at low visibility, false positives per hour, uncertainty calibration. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.51 51. Streaked frogfish (*Antennarius striatus*)

Archetype and observability. Streaked frogfish is treated here as a *camouflage* corpus problem in tropical/subtropical benthic. The operational behavior most relevant to data collection is ambush predator, cryptic. A typical recognition cue is globose irregular body with appendages, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are sponges, algae, debris, other frogfish. Prioritize masks and context patches; include extreme color morphs and substrate-matching examples.

Recommended modalities. The preferred primary modality is high-resolution video/stills with segmentation masks and background negatives. A complementary secondary source is macro specimen photos, habitat context, eDNA for presence confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. actively mine false positives from natural substrate; include low contrast and partial visibility. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks, visibility fraction, occlusion, substrate/habitat, uncertainty flag, diagnostic-part visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include substrate structures, coral, algae, sponges, debris and related cryptic taxa. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/habitat held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, recall at low visibility, false positives per hour, uncertainty calibration. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.52 52. White-spotted puffer (*Arothron hispidus*)

Archetype and observability. White-spotted puffer is treated here as a *reef fine* corpus problem in tropical reef/lagoon. The operational behavior most relevant to data collection is slow swimmer, can inflate. A typical recognition cue is rounded puffer body with spots, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other Arothron puffers. Include inflated/non-inflated states and frontal views where lateral pattern cues disappear.

Recommended modalities. The preferred primary modality is high-resolution reef video/stills with taxonomic hierarchy. A complementary secondary source is BRUVS, citizen science, eDNA and expert voucher imagery. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. prioritize hard confusers and natural reef context; use hierarchy for uncertain fine-grained labels. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store species/genus/family, boxes/masks, orientation, life stage, color phase, occlusion, diagnostic-character visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include same-genus and same-family species, juveniles, color phases, partial/oblique views. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/dive/camera held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include top-1/top-k species, genus/family accuracy, long-tail macro-F1, calibration and site-held-out accuracy. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.53 53. Spotted seahorse (*Hippocampus kuda*)

Archetype and observability. Spotted seahorse is treated here as a *camouflage* corpus problem in seagrass, mangrove, reef. The operational behavior most relevant to data collection is slow, vertical posture, grasping. A typical recognition cue is small seahorse silhouette, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other Hippocampus, vegetation stems. Macro/close-range corpus should include pose and occlusion; species-level labels may require expert morphology unavailable at low resolution.

Recommended modalities. The preferred primary modality is high-resolution video/stills with segmentation masks and background negatives. A complementary secondary source is macro specimen photos, habitat context, eDNA for presence confirmation. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. actively mine false positives from natural substrate; include low contrast and partial visibility. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store instance masks, visibility fraction, occlusion, substrate/habitat, uncertainty flag, diagnostic-part visibility. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include substrate structures, coral, algae, sponges, debris and related cryptic taxa. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use site/habitat held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include mask IoU, recall at low visibility, false positives per hour, uncertainty calibration. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.54 54. Zebrafish (*Danio rerio*)

Archetype and observability. Zebrafish is treated here as a *model fish* corpus problem in freshwater, laboratory model. The operational behavior most relevant to data collection is small schooling fish. A typical recognition cue is small striped cyprinid, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are related Danio, juvenile cyprinids. Separate laboratory strain, fluorescent transgenic, and wild-type domains; avoid tank-ID leakage in behavior studies.

Recommended modalities. The preferred primary modality is controlled laboratory video with experiment metadata. A complementary secondary source is wild-type reference images and specialized microscopy where relevant. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. balance strains, sexes, developmental stages, camera rigs and experimental conditions. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store individual/strain, sex, stage, trial, tank, treatment, pose/track, behavior label. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other strains, reflections, tank hardware, neighboring fish. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use experiment/tank/individual held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include behavior/identity accuracy, cross-rig and cross-experiment generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.55 55. Japanese medaka (*Oryzias latipes*)

Archetype and observability. Japanese medaka is treated here as a *model fish* corpus problem in freshwater/brackish, laboratory model. The operational behavior most relevant to data collection is small schooling model fish. A typical recognition cue is small ricefish, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other Oryzias, guppies. Record strain, sex, developmental stage and imaging setup; experiment-level splits are more important than frame-level splits.

Recommended modalities. The preferred primary modality is controlled laboratory video with experiment metadata. A complementary secondary source is wild-type reference images and specialized microscopy where relevant. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. balance strains, sexes, developmental stages, camera rigs and experimental conditions. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store individual/strain, sex, stage, trial, tank, treatment, pose/track, behavior label. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other strains, reflections, tank hardware, neighboring fish. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use experiment/tank/individual held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include behavior/identity accuracy, cross-rig and cross-experiment generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.56 56. Three-spined stickleback (*Gasterosteus aculeatus*)

Archetype and observability. Three-spined stickleback is treated here as a *model fish* corpus problem in freshwater and coastal marine. The operational behavior most relevant to data collection is territorial/spawning behavior. A typical recognition cue is small fish with dorsal spines, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other sticklebacks, juvenile fishes. Include breeding coloration and sex; behavior corpora need individual identity and trial metadata.

Recommended modalities. The preferred primary modality is controlled laboratory video with experiment metadata. A complementary secondary source is wild-type reference images and specialized microscopy where relevant. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. balance strains, sexes, developmental stages, camera rigs and experimental conditions. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store individual/strain, sex, stage, trial, tank, treatment, pose/track, behavior label. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other strains, reflections, tank hardware, neighboring fish. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use experiment/tank/individual held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include behavior/identity accuracy, cross-rig and cross-experiment generalization. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.57 57. Great white shark (*Carcharodon carcharias*)

Archetype and observability. Great white shark is treated here as a *elasmobranch* corpus problem in coastal/oceanic marine. The operational behavior most relevant to data collection is large mobile predator. A typical recognition cue is robust lamnid shark, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are mako shark, other lamnids. Use multi-view body/fin annotations and individual-ID metadata; protect whole encounters in test sets.

Recommended modalities. The preferred primary modality is underwater video/stills with shape and individual pattern annotations. A complementary secondary source is aerial imagery, tagging context, eDNA and sonar. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. include dorsal/ventral/lateral views, range and occlusion; preserve rare encounters as provenance-rich gold data. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, species, body orientation, fin landmarks, sex/stage where visible, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include related sharks/rays, partial fins, silhouettes, swimmers and surface glare. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use individual/encounter/site held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species/open-set accuracy, individual re-ID, fin/landmark metrics and site-held-out performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.58 58. Tiger shark (*Galeocerdo cuvier*)

Archetype and observability. Tiger shark is treated here as a *elasmobranch* corpus problem in tropical/subtropical coastal-oceanic. The operational behavior most relevant to data collection is large roaming predator. A typical recognition cue is broad head, vertical body bars, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are bull shark, lemon shark, requiem sharks. Include adults where bars fade and juveniles where bars are strong; site-held-out evaluation is essential.

Recommended modalities. The preferred primary modality is underwater video/stills with shape and individual pattern annotations. A complementary secondary source is aerial imagery, tagging context, eDNA and sonar. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. include dorsal/ventral/lateral views, range and occlusion; preserve rare encounters as provenance-rich gold data. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, species, body orientation, fin landmarks, sex/stage where visible, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include related sharks/rays, partial fins, silhouettes, swimmers and surface glare. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use individual/encounter/site held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species/open-set accuracy, individual re-ID, fin/landmark metrics and site-held-out performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.59 59. Scalloped hammerhead (*Sphyrna lewini*)

Archetype and observability. Scalloped hammerhead is treated here as a *elasmobranch* corpus problem in tropical coastal/oceanic. The operational behavior most relevant to data collection is schools and migrations. A typical recognition cue is hammer-shaped cephalofoil, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other hammerheads. Top/head geometry is diagnostic; collect oblique/distant views and school scenes where individuals overlap.

Recommended modalities. The preferred primary modality is underwater video/stills with shape and individual pattern annotations. A complementary secondary source is aerial imagery, tagging context, eDNA and sonar. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. include dorsal/ventral/lateral views, range and occlusion; preserve rare encounters as provenance-rich gold data. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, species, body orientation, fin landmarks, sex/stage where visible, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include related sharks/rays, partial fins, silhouettes, swimmers and surface glare. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use individual/encounter/site held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species/open-set accuracy, individual re-ID, fin/landmark metrics and site-held-out performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.60 60. Giant manta ray (*Mobula birostris*)

Archetype and observability. Giant manta ray is treated here as a *elasmobranch* corpus problem in tropical/subtropical pelagic/coastal. The operational behavior most relevant to data collection is large filter feeder. A typical recognition cue is large ray with cephalic lobes, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are reef manta, devil rays. Ventral pattern images support individual re-ID; separate species ID from individual ID and record photo side.

Recommended modalities. The preferred primary modality is underwater video/stills with shape and individual pattern annotations. A complementary secondary source is aerial imagery, tagging context, eDNA and sonar. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. include dorsal/ventral/lateral views, range and occlusion; preserve rare encounters as provenance-rich gold data. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, species, body orientation, fin landmarks, sex/stage where visible, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include related sharks/rays, partial fins, silhouettes, swimmers and surface glare. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use individual/encounter/site held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species/open-set accuracy, individual re-ID, fin/landmark metrics and site-held-out performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.61 61. Bluespotted ribbontail ray (*Taeniura lymma*)

Archetype and observability. Bluespotted ribbontail ray is treated here as a *elasmobranch* corpus problem in Indo-Pacific reef/benthic. The operational behavior most relevant to data collection is benthic, often resting. A typical recognition cue is small ray with blue spots, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other stingrays. Top-down imagery and substrate variation dominate; include partially buried states.

Recommended modalities. The preferred primary modality is underwater video/stills with shape and individual pattern annotations. A complementary secondary source is aerial imagery, tagging context, eDNA and sonar. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. include dorsal/ventral/lateral views, range and occlusion; preserve rare encounters as provenance-rich gold data. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, species, body orientation, fin landmarks, sex/stage where visible, individual ID when known. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include related sharks/rays, partial fins, silhouettes, swimmers and surface glare. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use individual/encounter/site held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include species/open-set accuracy, individual re-ID, fin/landmark metrics and site-held-out performance. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.62 62. Greenland shark (*Somniosus microcephalus*)

Archetype and observability. Greenland shark is treated here as a *deep sea* corpus problem in Arctic/North Atlantic deep cold water. The operational behavior most relevant to data collection is slow, deep-living. A typical recognition cue is large dark sleeper shark, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are other sleeper sharks, sixgill sharks. Expect sparse data: prioritize expert-verified events, ROV metadata and open-set evaluation rather than arbitrary sample targets.

Recommended modalities. The preferred primary modality is ROV/AUV imagery with synchronized depth, vehicle, lighting and navigation metadata. A complementary secondary source is specimen/voucher photographs, eDNA, sonar where available. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. preserve raw low-light data; cover multiple vehicles/lighting geometries and rare encounter contexts. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, taxonomic confidence, depth, range, lighting state, life stage, specimen link. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other deep-sea fishes, shadows, marine snow, vehicle artifacts. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use dive/expedition/seamount held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include open-set metrics, taxonomic hierarchical accuracy, calibration and expedition-held-out recall. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.63 63. West Indian Ocean coelacanth (*Latimeria chalumnae*)

Archetype and observability. West Indian Ocean coelacanth is treated here as a *deep sea* corpus problem in deep rocky slopes/caves. The operational behavior most relevant to data collection is slow, rare, nocturnal. A typical recognition cue is lobed fins, distinctive deep-bodied silhouette, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are large groupers, other deep fishes. Rare-species protocol: retain every provenance field, never split frames from one dive across sets, and use taxonomic hierarchy/novelty detection.

Recommended modalities. The preferred primary modality is ROV/AUV imagery with synchronized depth, vehicle, lighting and navigation metadata. A complementary secondary source is specimen/voucher photographs, eDNA, sonar where available. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. preserve raw low-light data; cover multiple vehicles/lighting geometries and rare encounter contexts. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, taxonomic confidence, depth, range, lighting state, life stage, specimen link. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other deep-sea fishes, shadows, marine snow, vehicle artifacts. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use dive/expedition/seamount held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include open-set metrics, taxonomic hierarchical accuracy, calibration and expedition-held-out recall. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

20.64 64. Sloane's viperfish (*Chauliodus sloani*)

Archetype and observability. Sloane's viperfish is treated here as a *deep sea* corpus problem in mesopelagic/deep ocean. The operational behavior most relevant to data collection is vertical migrator. A typical recognition cue is slender dark fish with large teeth, but the corpus must prevent a model from depending on a single iconic cue or background. The most important deliberate confusers for the first collection campaign are dragonfishes, lancetfish juveniles. Low-light ROV specimens and preserved-specimen images must be separate domains; illumination normalization should not erase photophore cues.

Recommended modalities. The preferred primary modality is ROV/AUV imagery with synchronized depth, vehicle, lighting and navigation metadata. A complementary secondary source is specimen/voucher photographs, eDNA, sonar where available. The secondary modality should be linked at the event or site level rather than merged as if every observation were perfectly synchronized. For this species, the primary corpus should contain the native deployment signal before any aggressive enhancement so future models can revisit preprocessing choices.

Acquisition design. preserve raw low-light data; cover multiple vehicles/lighting geometries and rare encounter contexts. Use a stratified acquisition matrix with site, season, life stage, range/scale, visibility, orientation, sensor identity, and behavior. The first collection should explicitly reserve budget for rare-but-important conditions instead of waiting for them to appear by chance. At least one stratum should approximate the worst operational visibility or signal-to-noise condition. If the species is rare, prioritize additional sites and independent encounters over dense sampling of the same encounter.

Annotations and metadata. Store boxes/masks, taxonomic confidence, depth, range, lighting state, life stage, specimen link. Every label should include taxonomic confidence and the evidence used for species confirmation. Where diagnostic characters are not visible, retain a valid genus/family label or "unresolved" status instead of forcing species certainty. Minimum metadata should include event/recording ID, site, date/time, sensor, depth/range when available, life stage, observer/verifier, source license, and a split-group identifier.

Hard negatives and corpus balance. Include other deep-sea fishes, shadows, marine snow, vehicle artifacts. Balance at the level of independent events. Hard negatives should be curated from the same sites and sensors as positives so the model cannot solve the task from location or image style. For fine-grained confusers, collect paired examples under similar orientation and scale. Keep true empty/background intervals because false positives per hour or per survey distance are often more operationally meaningful than classification accuracy.

Splits and validation. Use dive/expedition/seamount held-out. A random frame split is not acceptable for a deployment claim. The evaluation suite should include open-set metrics, taxonomic hierarchical accuracy, calibration and expedition-held-out recall. Report per-stratum performance for visibility/range, life stage, and major confusers. A final open-set test should include non-target species not represented as explicit training classes. The release card should document the domains that remain uncovered so future data collection is directed by evidence rather than raw volume.

A Species-corpus planning worksheet

For each new species, complete the following worksheet before annotation begins.

Field	Required decision
Deployment decision	Presence, identity, count, biomass, length, behavior, health, individual ID, or community composition.
Primary sensor	RGB/stereo, BRUVS, ROV/AUV, imaging sonar, echosounder, hydrophone, eDNA, specimen imaging.
Independent event	Encounter, track, school, dive, call, sample, tank trial, specimen, cruise.
Confusers	Same genus/family, similar silhouette/color, habitat objects, sensor artifacts.
Coverage strata	Site, season, stage, sex/color phase, behavior, range, visibility, device, density.
Annotation	Class, hierarchy, box, mask, track, pose, call event, sequence, barcode, confidence.
Gold-core rule	Double annotation/adjudication rate and protected benchmark policy.
Split group	Site, encounter, individual, source, camera, cruise, farm, waterbody, lab batch.
Open-set policy	Unknown taxa, unresolved labels, confidence threshold, abstention.
Licensing	File-level source, contributor, license and redistribution constraints.
Release QA	Hashes, manifest schema, duplicate search, taxonomy version, changelog.

B Recommended annotation handbook rules

1. Annotate the animal actually visible, not what the annotator expects from site context.
2. Preserve a hierarchy and uncertainty; "unresolved grouper" is better than an unjustified species label.
3. Define how partial bodies, reflections, shadows and images on screens are treated.
4. For schools, decide whether the project needs group regions, individual instances, density labels, or a mixture.
5. Define minimum resolvable size/visibility for species-level annotation and a separate "fish present, species unresolved" label.
6. In video, keep object IDs stable through short occlusions and specify the gap after which a new track begins.
7. Never assign adjacent frames from one track to different train/validation/test partitions.
8. Record whether the annotator used auxiliary evidence such as a second camera, specimen catch, eDNA, tag, or acoustic localization.
9. Audit a stratified sample by species and difficulty, not only a global random sample.
10. Log all guideline changes and maintain label-version provenance.

C Example JSONL record


```
{
  "event_id": "reef03_dive17_track0042",
  "file": "reef03/dive17/frame_003812.jpg",
  "scientific_name": "Chaetodon capistratus",
  "taxon_id": "local:taxon:01842",
  "family": "Chaetodontidae",
  "genus": "Chaetodon",
  "species": "capistratus",
```

```
  "label_rank": "species",
  "label_confidence": 0.96,
  "annotation": {"bbox": [811, 422, 126, 84], "track_id": 42},
  "life_stage": "adult_or_subadult",
  "orientation": "left_lateral_oblique",
  "occlusion_fraction": 0.15,
  "site_id": "reef03",
  "depth_m": 12.4,
  "visibility_m": 9.0,
  "sensor_model": "camera_A",
  "parent_recording_id": "dive17",
  "split_group": "reef03_dive17",
  "license": "project-defined",
  "annotation_version": "2.1"
}
```


D Recommended benchmark reporting table

A release paper should report the following per species and macro-averaged across species: independent training events, annotated instances, sites, devices, life stages, major confusers, in-domain performance, unseen-site performance, open-set false positive rate, calibration error, and annotation agreement. Counting/tracking systems should additionally report count error, track fragmentation and false tracks per unit time. eDNA systems should report marker/reference coverage, controls, replicate concordance, taxonomic assignment thresholds and unresolved assignments. Passive-acoustic systems should report verified calls by behavior/site/individual when known and false detections per recording hour.

E Research agenda

The next generation of fish corpora should move from static image collections toward provenance-rich multimodal benchmarks. Priority areas include: (1) taxonomically hierarchical labels with open-set evaluation; (2) standardized event-based splits; (3) cross-sensor domain benchmarks for cameras and sonar; (4) large verified passive-acoustic reference libraries; (5) voucher-linked eDNA reference coverage; (6) semi-automatic annotation with measured human correction; (7) data cards that expose geographic and life-stage gaps; (8) continual-learning evaluation when new species are added; and (9) privacy, licensing and Indigenous/community governance for sensitive locations and biodiversity records.

The most important conceptual shift is from asking "how many images per species?" to asking "which independent observations are missing for this species and deployment?" A corpus becomes scientifically useful when its structure makes that question answerable.

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